A Casimir–Secular Electroweak Mass Ratio and Its String/Kaluza–Klein Completion Problem

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Abstract

We formulate the Rivero–de Vries electroweak observation as a two-branch secular problem built from a spin-Casimir variable $J = s(s+1)$. The positive root of $x^2+Jx-J=0$, evaluated at $J=3/4$ and $J=2$, gives $1-x_+(3/4)/x_+(2) = 0.22310132\ldots$, close to the weak angle defined by the W/Z pole-mass ratio. The paper treats this agreement as a pole-spectrum statement and separates it from running weak mixing angles. We present the algebraic construction, the electroweak projective interpretation in which the Higgs vacuum scale is radial, and the analytical tasks required for a dynamical derivation. String, brane, Kaluza–Klein, and G_2-holonomy ingredients are organized as possible mechanisms for the secular equation rather than as independent assumptions. The central open calculation is the derivation of the ordered assignment $(J_W, J_Z) = (3/4, 2)$ from the gauge-Higgs sector or from a compactification boundary problem.

I. INTRODUCTION

The electroweak sector contains a dimensionless mass ratio,

$$s^2_{\mathrm{pole}} = 1 - \frac{M^2_{W,\mathrm{pole}}}{M^2_{Z,\mathrm{pole}}}, \tag{1}$$

which is fixed experimentally by the pole positions of the charged and neutral weak vector bosons. In the Standard Model at tree level the same ratio is written as an angular datum in coupling space, while the Higgs vacuum expectation value fixes the common radial scale of the vector masses. This separation between projective and radial data gives the working language of the present paper.

The Rivero–de Vries observation starts from a mass operator built from Poincare Casimir data and produces a dimensionless two-branch equation. In the normalized notation used below, the equation is

$$x^2 + Jx - J = 0, \qquad J = s(s+1). \tag{2}$$

The positive branch gives

$$s^2_{\mathrm{dV}} = 1 - \frac{x_+(3/4)}{x_+(2)} = 0.22310132\ldots. \tag{3}$$

The original note emphasized the proximity of this number to the mass-shell weak angle and motivated the operator through Casimir invariants and high-spin behavior [1]. The present manuscript reorganizes the point as a pole-spectrum statement and asks what physics can turn Eq. (2) into an electroweak calculation.

The construction has three layers. The first layer is algebraic: a two-channel secular equation has roots whose ratio is fixed by the Casimir variable. The second layer is electroweak: the W/Z mass ratio is a projective datum along the broken-to-unbroken ray of the gauge-Higgs system. The third layer is dynamical: strings, branes, Kaluza–Klein boundary conditions, or singular G_2 geometries may generate the secular equation or its branch structure.

The status of the argument is deliberately stratified. The algebraic calculation is explicit. The numerical comparison is reproducible. The pole-mass scheme must be source-audited. The assignment of $J = 3/4$ to the charged W comparison and $J = 2$ to the neutral Z comparison is the central open analytical calculation unless it is derived in Sec. IV. The negative branch is retained because it may encode the scalar/order-parameter side of electroweak symmetry breaking.

The paper is organized as follows. Section II gives the secular operator and its roots. Section III defines the pole observable. Section IV formulates the projective electroweak interpretation. Section V records the negative branch and its possible Higgs-scale role. Section VI states the string/Kaluza–Klein/G_2 completion problem. Sections VII and VIII delimit the flavor and global-gauge-group issues. Section IX gives the status of the analytical program and falsifiable tests.

Section status. This introduction states the program and labels the central unproved step.

II. CASIMIR-SECULAR MASS OPERATOR

Let $J = s(s+1)$ be the spin-Casimir eigenvalue. The normalized two-channel operator used in this manuscript is

$$\frac{Q(J)}{\mu^2} = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \tag{4}$$

Its characteristic equation is

$$\det\left(xI - \frac{Q(J)}{\mu^2}\right) = x^2 + Jx - J = 0. \tag{5}$$

The two roots are

$$x_\pm(J) = \frac{-J \pm \sqrt{J^2+4J}}{2}. \tag{6}$$

They obey

$$x_+(J) + x_-(J) = -J, \qquad x_+(J)x_-(J) = -J. \tag{7}$$

For $J > 0$, the positive branch is positive and the negative branch is negative.

The electroweak number is obtained by sampling the positive branch at the two Casimir values

$$J_{1/2} = \frac{3}{4}, \qquad J_1 = 2. \tag{8}$$

This gives

$$s^2_{\rm dV} = 1 - \frac{x_+(3/4)}{x_+(2)}. \tag{9}$$

The expression is dimensionless; the overall scale μ cancels. The cancellation is essential for an electroweak interpretation because Eq. (9) can then be compared with a projective W/Z mass ratio.

The relation to the original Rivero operator is obtained by restoring the Poincare Casimir eigenvalues $C_1 = m^2$ and $C_2 = -m^2 s(s+1)$. In the original construction, a mass-squared operator is formed from combinations of Casimir invariants, with a high-spin condition designed to preserve the asymptotic mass behavior relevant for Regge trajectories [1]. Equation (4) is the reduced dimensionless representative of that branch problem.

A complete derivation should state the minimal assumptions that select Eq. (4). The current minimal list is: dimensionally correct dependence on the Poincare Casimirs, exclusion of inverse Casimirs, a two-branch square-root structure, and preservation of the high-spin asymptotic datum. A trace condition can be used as an alternative route. The uniqueness of the construction under these assumptions remains a theorem to be written with explicit hypotheses.

Section status. The eigenvalue algebra is derived. The uniqueness of the operator under the stated constraints is an open analytical subtask.

III. POLE OBSERVABLE AND ELECTROWEAK SCHEME SEPARATION

The observable used for comparison is the pole-mass ratio

$$s^2_{\rm pole} = 1 - \frac{M^2_{W,\rm pole}}{M^2_{Z,\rm pole}}. \tag{10}$$

This ratio is distinct from a running $\overline{\text{MS}}$ weak angle and from effective weak mixing angles extracted from neutral-current asymmetries. The manuscript therefore treats Eq. (10) as the primary target and later relates other schemes to it.

A vector-boson pole is defined by the complex pole of the propagator,

$$s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \tag{11}$$

When experimental inputs are quoted in a Breit–Wigner convention, the conversion to pole parameters must be performed before using Eq. (10). This is one of the numerical audit tasks of the project.

The project seed comparison uses

$$M_W = M_Z\sqrt{1 - s_{\text{dV}}^2}. \tag{12}$$

For the seed value $M_Z = 91.1880$ GeV, this gives $M_W = 80.3748$ GeV. These numbers are placeholders inherited from the working draft and must be audited against the current electroweak source set before final submission. The calculation script in Appendix B reproduces the value.

The Standard Model parameter literature distinguishes several vacuum and renormalization conventions. In particular, pure $\overline{\text{MS}}$ treatments may define the Higgs vacuum expectation value as the minimum of the loop-corrected effective potential in Landau gauge, eliminating tadpole diagrams at the cost of gauge-choice dependence in intermediate parameters [9]. Such scheme details are relevant for comparing a mass-ratio statement with Lagrangian parameters, while Eq. (10) remains the pole-spectrum target.

Section status. The observable is defined. The numerical input table requires a current electroweak source audit before final claims.

IV. ELECTROWEAK EMBEDDING AS A PROJECTIVE DATUM

The gauge-Higgs sector gives the tree-level vector masses

$$M_W^2 = \frac{g^2v^2}{4}, \qquad M_Z^2 = \frac{(g^2+g'^2)v^2}{4}, \qquad M_\gamma^2 = 0. \tag{13}$$

The ratio is

$$1 - \frac{M_W^2}{M_Z^2} = \frac{g'^2}{g^2+g'^2}. \tag{14}$$

Thus the vacuum scale v fixes the radial size of the massive vector pair, while the quotient fixes a projective direction in the two-coupling plane.

The DeVries proposal is read in the same projective language:

$$\frac{M_W^2}{M_Z^2} \quad \longleftrightarrow \quad \frac{x_+(3/4)}{x_+(2)}. \tag{15}$$

The common scale is then supplied by the electroweak vacuum or, in a deeper model, by the compactification or brane scale that generates the effective gauge-Higgs sector.

The unbroken limit is the ray

$$v \to 0 \quad \text{with} \quad \frac{g'^2}{g^2 + g'^2} \text{ fixed.} \tag{16}$$

Along this ray, M_W and M_Z vanish together and the photon remains massless. This is the electroweak deformation relevant to the construction. The projective datum remains meaningful as the limiting direction even when the radial mass scale is taken to zero.

The central analytical question is the ordered assignment

$$(J_W, J_Z) = \left(\frac{3}{4}, 2\right). \tag{17}$$

The working interpretation is that these entries encode electroweak representation data associated with the symmetry-breaking direction. The $J = 3/4$ datum is naturally associated with a doublet-like $T = 1/2$ object, while $J = 2$ is naturally associated with an adjoint-like $T = 1$ object. This statement is a candidate route rather than a completed derivation. A completed derivation must start from the gauge-Higgs Lagrangian or from a compactification boundary problem and produce Eq. (17) without fitting to the observed masses.

Section status. The projective interpretation follows from the electroweak mass matrix. The representation-theoretic origin of Eq. (17) remains the primary open calculation.

V. NEGATIVE BRANCH AND THE SCALAR SIDE OF THE PROBLEM

The negative branch in Eq. (6) is fixed by the same secular equation as the positive branch. It should therefore be retained in the analytical structure even if the W/Z ratio uses only x_+. The root identities imply

$$x_-(J) = -J - x_+(J) = -\frac{J}{x_+(J)}. \tag{18}$$

Any physical use of the positive branch automatically carries a partner branch.

The Higgs potential can be written as

$$V(H) = m^2 H^\dagger H + \lambda (H^\dagger H)^2, \qquad m^2 < 0 \tag{19}$$

in the broken phase. The sign of the quadratic term makes the negative branch tempting as an order-parameter datum. A valid identification must be gauge-invariant and scheme-controlled. Possible targets include m^2, λv^2, the Higgs pole mass, or an eigenvalue in a compactification problem.

The branch values associated with $J = 3/4$ and $J = 2$ generate definite dimensionless numbers. Multiplying them by a scale fitted to the vector masses gives scales near the electroweak/Higgs range. This observation is a prompt for analysis. The manuscript should not promote it to a prediction until a scalar-sector map is derived.

A useful criterion is the following. If the same two-channel determinant produces both the vector pole ratio and the scalar instability, then the determinant should appear in an effective action, boundary condition, or self-energy equation. The next calculation is to write such a determinant explicitly and identify which field or mode occupies the second channel.

Section status. The negative branch algebra is exact. Its Higgs interpretation is conjectural and requires a dynamical map.

VI. STRING, KALUZA–KLEIN, AND G_2 COMPLETION PROBLEM

The secular equation has the form of a low-level branch splitting that preserves a high-spin datum. This is the reason string and Regge language enter the project. In string theory, higher-spin excitations organize into Regge trajectories, and the asymptotic slope is controlled by the string tension. The Rivero construction was motivated in part by preserving such an asymptotic datum while modifying low-spin masses [1, 2].

A Kaluza–Klein or brane derivation would have to produce the determinant

$$\det \begin{pmatrix} x & -\sqrt{J} \\ -\sqrt{J} & x + J \end{pmatrix} = 0. \tag{20}$$

This can arise from a two-channel boundary problem, a mixing of brane-localized and bulk modes, or an effective orbit problem. The required output is not merely a qualitative similarity; it is the exact J-dependence in Eq. (20).

Extra-dimensional electroweak models show that boundary conditions can break gauge symmetry and determine vector spectra. Higgsless constructions on intervals provide a controlled language for relating boundary data, vector masses, fermion localization, and precision observables [6]. In the present project, such models serve as templates for deriving a mass-ratio constraint from boundary conditions.

M-theory on G_2-holonomy manifolds supplies a separate completion map. Smooth compact G_2 compactifications give four-dimensional minimal supersymmetry but do not by themselves yield realistic charged chiral matter in the large smooth limit. Singularities are required for nonabelian gauge sectors and chiral fermions [11–13]. Local Higgs-bundle descriptions over associative three-cycles give a modern way to organize gauge sectors, localized matter, and singular transitions [14].

The completion problem can therefore be stated as a table of mechanisms.

Mechanism	What it can supply	Required DeVries test
Regge/string tower	high-spin asymptotics	preserve slope while producing Eq. (20)
Interval/brane BC	vector mass spectra	derive $J = 3/4, 2$ samples from boundary data
KK compactification	mode eigenvalues	produce two-channel branch determinant
Singular G_2	gauge/chiral sectors	localize the electroweak operator geometrically
Higgs bundle	matter/mass localization	identify the two branches as spectral-cover data

Section status. This section states completion mechanisms and tests. No dynamical derivation is claimed here.

VII. FLAVOR BOUNDARY AND REPRESENTATION ALGEBRA

The electroweak mass-ratio problem is separated from the generation problem. The working boundary condition is that generations are not assumed to arise exactly from compactification topology. A separate SO(32)-flavor interpretation may organize fermion content and endpoints, and it may or may not be compatible with a compactification geometry.

Representation-theoretic unification remains useful because it constrains how Standard Model fermions can sit inside larger groups. The algebraic GUT literature gives clean embeddings through SU(5), Spin(10), Pati–Salam, and E6 representation theory [17, 18].

These embeddings can guide the flavor and endpoint sector without determining the DeVries W/Z ratio by themselves.

A future compactification model must satisfy two independent constraints. First, it must reproduce the electroweak projective mass datum. Second, it must accommodate the observed chiral matter and flavor structure. The project does not identify these two constraints unless an explicit geometry or algebraic mechanism forces the identification.

Section status. This section is a boundary condition for the broader project. It prevents overclaiming about topology and generations.

VIII. GLOBAL FORM OF THE STANDARD MODEL GAUGE GROUP

The local Standard Model gauge algebra does not by itself fix the global gauge group. The global form may be written as

$$G_{\mathrm{SM}} = \frac{SU(3)_C \times SU(2)_W \times U(1)_Y}{\Gamma}, \qquad \Gamma \subset \mathbb{Z}_6. \tag{21}$$

The possibilities $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$ define different spectra of line operators and different topological sectors. Local correlation functions in flat spacetime do not fix this choice, while line operators and global probes can distinguish it [5, 15].

This global-form issue is relevant to any topological or compactification completion of the DeVries construction. A model that derives the electroweak secular equation should also specify which line operators exist and which quotient Γ is realized. Recent work proposes topological transport coefficients that can distinguish the global form in analogy with fractional quantum Hall response [16]. Such tests are not inputs to the mass ratio, but they constrain the class of completions.

Section status. The global-form appendix is source-supported and functions as a completion constraint.

IX. STATUS OF THE ANALYTICAL PROGRAM AND FALSIFIABLE TESTS

The construction yields a precise number from a precise branch quotient. The electroweak comparison is meaningful only after choosing the pole-mass observable. The project advances when the following calculations are completed.

1. Derive the ordered assignment $(J_W, J_Z) = (3/4, 2)$ from the gauge-Higgs sector or from a boundary/compactification problem.

2. Derive the two-channel determinant Eq. (20) from a dynamical model.

3. Determine whether the negative branch maps to the Higgs instability or to another auxiliary sector.

4. Update the pole-mass comparison with current source-audited inputs and uncertainty propagation.

5. Specify the global form of the gauge group in any proposed topological completion.

The falsifiable tests are correspondingly direct. A source-audited pole-mass ratio that decisively departs from Eq. (9) would weaken the central observation. A proof that no gauge-Higgs or boundary construction can produce Eq. (17) under the required assumptions would block the electroweak interpretation. A dynamical derivation that produces a different ordered pair or a different branch quotient would redirect the project toward a broader class of secular mass relations.

Section status. This section states the program's current logical dependencies.

Appendix A: Algebraic details

For completeness, we record the elementary algebra. From Eq. (5),

$$x_\pm(J) = \frac{-J \pm \sqrt{J(J+4)}}{2}. \tag{A1}$$

The positive branch has the large-J expansion

$$x_+(J) = 1 - \frac{1}{J} + \frac{2}{J^2} + O(J^{-3}), \tag{A2}$$

while

$$x_-(J) = -J - 1 + \frac{1}{J} + O(J^{-2}). \tag{A3}$$

Thus the positive branch approaches a fixed asymptotic value while the negative branch carries the large negative partner required by the trace.

For $J = 3/4$,

$$x_+\left(\frac{3}{4}\right) = \frac{-3/4 + \sqrt{57}/4}{2} = \frac{\sqrt{57} - 3}{8}. \tag{A4}$$

For $J = 2$,

$$x_+(2) = \frac{-2 + \sqrt{12}}{2} = \sqrt{3} - 1. \tag{A5}$$

Therefore

$$s^2_{\mathrm{dV}} = 1 - \frac{\sqrt{57} - 3}{8(\sqrt{3} - 1)}. \tag{A6}$$

Appendix B: Numerical reproducibility

The script ‘calculations/devries_spectrum.py’ computes the branch roots and the derived mass value from a supplied M_Z. The central functions are

```
x_plus(J) = (sqrt(J*J + 4*J) - J)/2
sin2_devries = 1 - x_plus(3/4)/x_plus(2)
MW = MZ * sqrt(1 - sin2_devries)
```

The test file ‘calculations/tests/test_devries_spectrum.py’ checks the root identities, the numerical value of s^2_{dV}, and the seed value obtained from $M_Z = 91.1880\,\mathrm{GeV}$.

Appendix C: Codex task map

The agentic workflow used for this manuscript is encoded in ‘codex_tasks/’. The tasks separate algebra, pole-mass scheme, electroweak interpretation, string/Kaluza–Klein/G_2 completion, flavor boundary, global-form appendix, and referee review. This separation is useful because the strongest numerical observation and the strongest conceptual gap are not the same object. The numerical observation is Eq. (9); the conceptual gap is the derivation of Eq. (17).

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